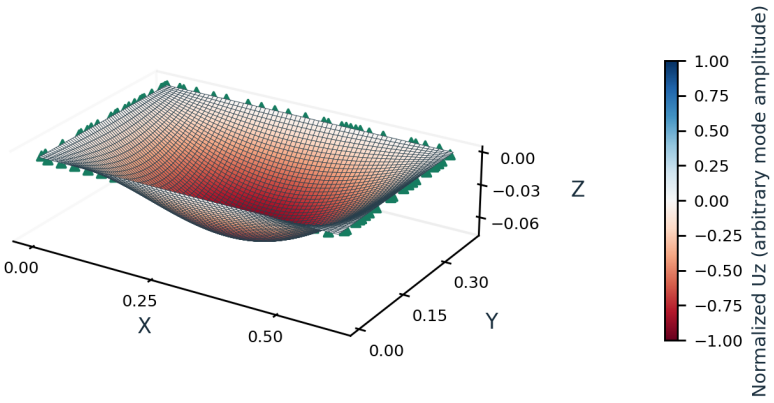
Symmetric cross-ply plate: Navier buckling

ACA03 | SOL 105 | Independent analytical equation verification

A simply supported symmetric cross-ply laminate under uniform compression tests composite bending stiffness, prestress and the first six buckling roots against the finite-plate Navier solution.

Geometry / mesh	Plate 0.6 m by 0.4 m, thickness 2 mm; CQUAD4 meshes from 4x4 to 64x64.
Material / section	[0/90/90/0], four 0.5 mm plies. E1 = 135 GPa, E2 = 10 GPa, G12 = G13 = G23 = 5 GPa, nu12 = 0.3.
Boundary conditions	All four edges uz=0; bending rotations free. ux=0 at x=0; origin uy=0; drilling rotation constrained.
Applied load	Uniform reference compression Nx = 1000 N/m on x = 0.6 m, consistent nodal traction. Same constraints in preload and buckling.

Mode 1 | lambda=5.6495



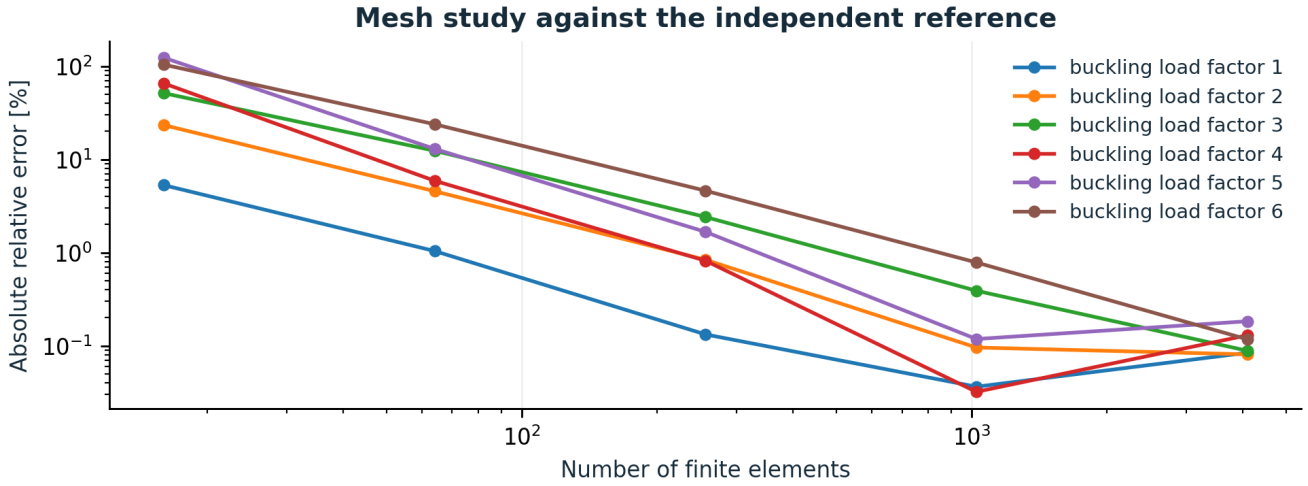
Shape amplified x0.072; grey = original. Eigenvector sign and amplitude are arbitrary.

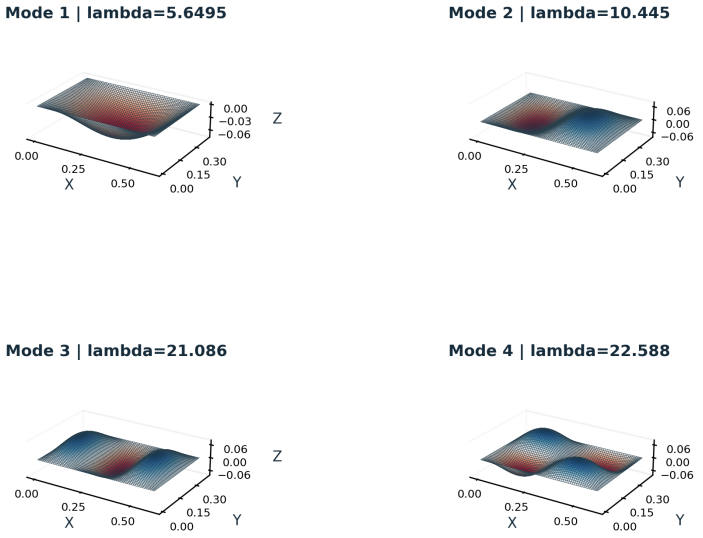
Actual solver output. Deformation is amplified for visibility; mode vectors have arbitrary sign and amplitude.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
buckling load factor 1 [dimensionless]	5.6495	5.6543	0.084284%	PASS
buckling load factor 2 [dimensionless]	10.445	10.453	0.080958%	PASS
buckling load factor 3 [dimensionless]	21.086	21.104	0.08859%	PASS
buckling load factor 4 [dimensionless]	22.588	22.617	0.1297%	PASS
buckling load factor 5 [dimensionless]	28.243	28.295	0.18339%	PASS
buckling load factor 6 [dimensionless]	36.321	36.363	0.11759%	PASS

Evidence, convergence and interpretation





Independently normalized eigenvectors; sign and amplitude arbitrary. Grey = original.

First available modes. Red/blue indicate opposite signed components; each mode is normalized independently.

Independent reference equations

D_ij=sum_k Qbar_ij(k)*(z_k^3-z_(k-1)^3)/3

alpha=m*pi/a; beta=n*pi/b

H=D11*alpha^4+2*(D12+2D66)*alpha^2*beta^2+D22*beta^4

lambda_mn=H/(Nx*alpha^2)

All first six finest-mesh factors agree within 0.184%; smallest transverse mode-shape MAC = 0.99999847.

All coarse errors remain tabulated;64x64 refinement improves the worst error from0.7851% at32x32.

Scope: Navier theory assumes a thin, flat, symmetric specially orthotropic plate. Transverse-shear flexibility and discretization distinguish the FE model from this idealization.

Reference and reproduction

Weaver and Nemeth, Improved Design Formulae for Buckling of Orthotropic Plates under Combined Loading (2008) https://ntrs.nasa.gov/citations/20080015745

Nachiketa Tiwari / IIT Kanpur: Mechanics of Laminated Composite Structures, Lecture37 Buckling of Plates https://archive.nptel.ac.in/content/storage2/courses/112104161/L37.pdf

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997cce40 | Full precision values: results.json / metrics.csv